

# Equation Sheet: Electrostatics

Intro Physics II • PHYS 230 • Knight ch. 22–25

**fundamental** = know cold (your starting points); **derived** = a line or two from the fundamentals.

## Symbols and constants

|                                                                                                                                                           |                                                           |                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|------------------------------|
| $q, Q$ charge (C)                                                                                                                                         | $\Phi$ electric flux (N·m <sup>2</sup> /C)                | $U$ potential energy (J)     |
| $\vec{E}$ electric field (N/C = V/m)                                                                                                                      | $\lambda, \eta, \rho$ line / area / volume charge density | $V$ potential (V = J/C)      |
| $r, d$ distance / separation (m)                                                                                                                          | $A$ area                                                  | $e$ $1.60 \times 10^{-19}$ C |
| $\epsilon_0$ $8.85 \times 10^{-12}$ C <sup>2</sup> /(N·m <sup>2</sup> ), $\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9$ N·m <sup>2</sup> /C <sup>2</sup>   |                                                           |                              |
| $\vec{r}_{12}$ separation vector, <i>points from charge 1 to charge 2</i> : $\vec{r}_{12} = \vec{r}_2 - \vec{r}_1$ ; $\hat{r}_{12} = \vec{r}_{12}/r_{12}$ |                                                           |                              |

## 1. Coulomb's law & the electric field (Knight 22.4–22.5, 23.1–23.2)

### fundamental

Force between two point charges  
(along the line joining them; repulsive for like signs;  $F \propto 1/r^2$ )

$$F = \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_2|}{r_{12}^2}$$

Field — force per unit test charge

$$\vec{E} = \frac{\vec{F}_{\text{on } q}}{q}, \quad \vec{F} = q\vec{E}$$

Field of a point charge  
(away from +, toward −)

$$E = \frac{1}{4\pi\epsilon_0} \frac{|q|}{r^2}$$

Superposition (both are vector sums)

$$\vec{F}_{\text{net}} = \sum_i \vec{F}_i, \quad \vec{E}_{\text{net}} = \sum_i \vec{E}_i$$

### derived

Coulomb's law, full vector form  
( $\vec{r}_{12}$  points from charge 1 to charge 2; force *on* 2, *due to* 1)

$$\vec{F}_{1 \text{ on } 2} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}^2} \hat{r}_{12}$$

## 2. Continuous charge distributions (Knight 23.3–23.6)

### fundamental

Chop into pieces, add up (integrate)

$$\vec{E} = \int d\vec{E}, \quad dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2}$$

Charge densities

$$\lambda = \frac{Q}{L}, \quad \eta = \frac{Q}{A}, \quad \rho = \frac{Q}{V}$$

### derived

Infinite line of charge (distance  $r$ )

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

Infinite plane / sheet of charge

$$E = \frac{\eta}{2\epsilon_0} \quad (\text{uniform — no } r)$$

Ring of charge  $Q$ , radius  $R$ , on axis at  $z$

$$E = \frac{1}{4\pi\epsilon_0} \frac{Qz}{(z^2 + R^2)^{3/2}}$$

Disk of charge  $Q$ , radius  $R$ , on axis at  $z$   
( $\eta = Q/\pi R^2$ ;  $R \rightarrow \infty$  recovers the infinite sheet;  
 $z \gg R$  recovers a point charge)

$$E = \frac{\eta}{2\epsilon_0} \left( 1 - \frac{z}{\sqrt{z^2 + R^2}} \right)$$

### 3. Electric flux & Gauss's law (Knight 24.1–24.3)

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#### fundamental

Electric flux through a surface

$$\Phi = \oint \vec{E} \cdot d\vec{A}$$

Gauss's law

(any closed surface; only the *enclosed* charge matters)

$$\Phi = \frac{Q_{\text{enc}}}{\epsilon_0}$$

#### derived

Uniform field through a flat area

$$\Phi = EA \cos \theta \quad (\theta \text{ from the normal})$$

Charge *outside* a closed surface

$$\Phi_{\text{net}} = 0$$

Sphere / spherical shell, total charge  $Q$   
( $r > R$ )

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \quad (\text{same as a point charge})$$

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### 4. Conductors in equilibrium (Knight 24.4–24.6)

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#### fundamental

Field inside the conducting material

$$\vec{E} = 0$$

Excess charge

all on the outer surface

Field at the outer surface

$$\perp \text{ to the surface,} \quad E = \frac{\eta}{\epsilon_0}$$

#### derived

Empty cavity inside a conductor

$$\vec{E} = 0 \text{ in the cavity} \quad (\text{Faraday shielding})$$

Charge  $+q$  placed in the cavity

$-q$  on the inner wall,  $+q$  on the outer surface

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### 5. Electric potential energy & potential (Knight 25.1–25.7)

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#### fundamental

Two point charges

(keep the signs;  $U \rightarrow 0$  as  $r_{12} \rightarrow \infty$ ; one power of  $r$ )

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}}$$

Energy conservation

$$K_i + U_i = K_f + U_f, \quad W_{\text{elec}} = -\Delta U$$

Potential — energy per unit charge

$$V = \frac{U}{q}, \quad U = qV \quad (1 \text{ V} = 1 \text{ J/C})$$

Point charge

(scalar; keep the sign;  $V \rightarrow 0$  as  $r \rightarrow \infty$ )

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}, \quad V_{\text{net}} = \sum_i \frac{1}{4\pi\epsilon_0} \frac{q_i}{r_i}$$

#### derived

Three or more charges: sum over every *pair*

(the work to assemble the whole arrangement — order doesn't matter)

$$U_{\text{total}} = \frac{1}{4\pi\epsilon_0} \sum_{\text{pairs } (i,j)} \frac{q_i q_j}{r_{ij}}$$

Equipotentials

$$\perp \text{ field lines; } \vec{E} \text{ points toward decreasing } V$$

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Companion simulations at [hoppese.github.io/Intro-physics-sims/](https://hoppese.github.io/Intro-physics-sims/): electroscope, e-fields-conductors-and-insulators, electrostatics-explorer, field-superposition-explorer, discrete-charge-builder-3d, gauss-flux, gauss-applications, assembling-charges, pe-vs-separation, equipotential-explorer.